

Per-Patient Bayesian Calibration of a Coupled α -Synuclein Aggregation–Neuron Death ODE in Parkinson’s Disease: Multi-Channel Identifiability from Longitudinal DaT-SPECT, CSF α -Synuclein, and Plasma GFAP in 2,118 PPMI Patients

Blair D. Dupre, Ph.D.¹

¹Department of Biomedical Engineering, University of North Dakota, Grand Forks, ND, USA

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

Quantitative systems pharmacology of Parkinson’s disease (PD) is limited to population-level or *in vitro* α -synuclein ODE calibrations [12, 27, 33]; no published study has calibrated a compartmental α -synuclein aggregation–neuron death ODE to individual PD patients from longitudinal neuroimaging with formal structural and practical identifiability analysis.

WHAT QUESTION DID THIS STUDY ADDRESS?

Can a coupled $[M, O, F, N]$ ODE— α -synuclein monomer, oligomer, fibril, and dopaminergic neuron count—be calibrated per patient from PPMI data under slow–fast timescale collapse, and does adding CSF α -synuclein and multi-channel biomarkers (GFAP, NfL, Amprion SAA TTT, neuron-derived extracellular-vesicle α -syn) break the k_n – α_{tox} identifiability ridge?

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

We deliver the first per-patient Bayesian posterior (importance sampling, 50 000 draws per subject, $n=304$) over $(k_n, \alpha_{\text{tox}})$ in PD. Adding CSF α -synuclein reduces the k_n – α_{tox} correlation

from -0.240 to -0.113 and tightens individual k_n posteriors by 29% cohort-wide (49% on high-information subset). A 5-channel SAEM extension on 2,118 patients delivers a matched-cohort GFAP ablation that narrows $\sigma_{\log k_n}$ by $37\times$ and sharpens the full-ODE Fisher-Information-Matrix condition number from $\kappa \approx 10^{12}$ (steady-state) to $\kappa = 19.86$ (transient). Leave-one-out forward-simulation coverage is 97.9% at 95% prediction interval across 1,940 held-out scans.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

Per-patient posteriors enable falsifiable counterfactuals: a calibrated prasinezumab simulation ($\eta_{\text{abx}} = 0.15$) predicts 17.6% slowing of SBR decline, an effect size below detection at PASADENA’s 52-week sample size and consistent with its null imaging endpoint [23]. The per-patient $T_{\text{tox}} = \alpha_{\text{tox}} k_n M_{\text{ss}}^2 / (k_{\text{conv}} + k_{\text{clear}}^O)$ quantifies the minimum detectable effect at any sample size, supporting enrichment-based adaptive designs for disease-modifying antibody trials.

Abstract

Background. Parkinson’s disease mechanistic models have not been calibrated per patient with formal identifiability analysis, limiting counterfactual inference.

Methods. We calibrated a four-state $[M, O, F, N]$ coupled α -synuclein–neuron-death ODE to PPMI longitudinal DaT-SPECT via importance sampling under slow–fast timescale collapse ($n=304$ Wave A), extended to joint DaT + CSF α -synuclein ($n=277$), and then to population-level SAEM on five observation channels (DaT + plasma GFAP + serum NfL + Amprion SAA TTT + neuron-derived-vesicle α -syn) across $n=2,118$. Structural identifiability was verified with the Julia package `StructuralIdentifiability.jl`; practical identifiability was assessed by profile-likelihood analysis and Fisher-information-matrix (FIM) conditioning.

Results. Cohort-median implied neuron loss was 3.29%/yr (SBR-only) and 2.93%/yr (SBR+CSF), within the Fearnley & Lees canonical 2–5%/yr range. CSF addition reduced the k_n – α_{tox} posterior correlation from -0.240 to -0.113 and tightened k_n by 29% cohort-wide (49% on HIGH-INFO patients). The 5-channel SAEM sharpened the full-ODE FIM condition number to $\kappa = 19.86$ (versus $\kappa \approx 10^{12}$ steady-state), with a matched-cohort GFAP ablation contributing a $37\times$ reduction

in $\sigma_{\log k_n}$. Leave-one-out forward coverage was 97.9% at 95% prediction interval (1,940 scans). The prasinezumab counterfactual predicted 17.6% SBR-decline slowing, below PASADENA’s detection threshold—consistent with its null imaging endpoint.

Discussion. Per-patient Bayesian posteriors over a compartmental PD ODE are identifiable when paired with transient-state multi-channel observations, and deliver falsifiable counterfactuals for disease-modifying-therapy trial design.

Keywords: Parkinson’s disease; α -synuclein; ODE calibration; identifiability; importance sampling; SAEM; DaT-SPECT; GFAP; counterfactual simulation

Introduction

Parkinson’s disease (PD) is driven at the molecular level by α -synuclein aggregation into toxic oligomeric and fibrillar species that precipitate progressive nigrostriatal dopaminergic neuron loss [4]. The Neuronal α -Synuclein Disease Integrated Staging System (NSD-ISS) [30] provides a biological classification for PD using seed amplification assay positivity (S anchor) and dopaminergic deficit on DaT-SPECT (D anchor), but does not specify a mechanistic model of *why* patients transition between stages or *when* transitions will occur. Several groups have proposed frameworks for mechanistic PD models [2, 6], but to date no published study has calibrated a compartmental α -synuclein ODE to individual patients from longitudinal neuroimaging with formal identifiability analysis. Adjacent recent work includes Hemedan et al. [11] (governed Bayesian twin on PPMI clinical scores without a compartmental kinetic model), Ivanova & Karelina [12] (mouse population QSP), and Righetti et al. [27] (*in vitro* α -synuclein ODE). None combines (i) a compartmental $[M, O, F, N]$ state space with (ii) per-patient Bayesian calibration from (iii) longitudinal DaT-SPECT with (iv) formal structural and practical identifiability analysis and (v) a second biomarker observable for degeneracy breaking.

Our primary question was whether a per-patient Bayesian posterior over α -synuclein aggregation nucleation (k_n) and neurotoxicity coupling (α_{tox}) can be identified from longitudinal DaT-SPECT and CSF α -synuclein in the Parkinson’s Progression Markers Initiative (PPMI), and whether it delivers counterfactual predictions consistent with the PASADENA null imaging endpoint [23]. A secondary question, addressed by a 5-channel extension (§), is whether adding plasma GFAP, serum NfL, Amprion SAA titre-time-to-threshold (TTT), and neuron-derived extracellular-vesicle (NEV) α -synuclein on a larger cohort ($n=2,118$) delivers additional identifiability gains via matched-cohort biomarker ablation.

Methods

Coupled ODE Framework (Variant B)

The model tracks α -synuclein monomer (M), oligomer (O), fibril (F), and neuron count (N):

$$\frac{dM}{dt} = k_{\text{prod}} - k_{\text{clear}}^M M - k_n M^2 - k_e M F \quad (1)$$

$$\frac{dO}{dt} = k_n M^2 - k_{\text{conv}} O - k_{\text{clear}}^O O \quad (2)$$

$$\frac{dF}{dt} = k_{\text{conv}} O - k_{\text{clear}}^F F \quad (3)$$

$$\frac{d \log(N/N_0)}{dt} = -(\alpha_{\text{tox}} O + k_{\text{age}}) \quad (4)$$

Equation (3) is the **Variant B mass-conserving formulation**, which removes a fragmentation-as-mass-creation error ($k_{\text{frag}} F$ in the naive single-state adaptation of the Cohen–Knowles framework [5, 15]) that caused $F \rightarrow 10^{75}$ nM at $t=4$ years in the initial implementation. The $\log-N$ transform (Eq. (4)) enforces neuron-count positivity by construction.

Under slow–fast timescale collapse (M, O, F equilibrate in hours; N relaxes over years), the SBR observation follows a closed-form exponential decay:

$$\text{SBR}(t) = \text{SBR}_0 \cdot \exp(-\gamma (\alpha_{\text{tox}} O_{\text{ss}} + k_{\text{age}}) t) \quad (5)$$

with $\gamma = 0.7$ and the toxicity flux composite $T_{\text{tox}} = \alpha_{\text{tox}} k_n M_{\text{ss}}^2 / (k_{\text{conv}} + k_{\text{clear}}^O)$ governing the SBR decline rate.

Structural and Practical Identifiability

`StructuralIdentifiability.jl` (v0.5.19), cross-validated with `SIAN.jl`, confirmed the reduced 2-parameter fit set $(k_n, \alpha_{\text{tox}})$ is globally identifiable from SBR alone [36]. Under slow–fast collapse, however, SBR depends only on the product T_{tox} , so individual k_n and α_{tox} are practically non-identifiable from SBR alone [10, 31]—the classical structural-vs-practical distinction [26]. Adding CSF α -synuclein as a second observable was motivated by this diagnosis. Full identifiability tables and profile-likelihood plots are provided as Supporting Information (SI §1).

Importance-Sampling Posterior

Single-chain NUTS sampling produced false-positive convergence ($\hat{R} < 1.05$) on prior-dominated posteriors [32]. We replaced NUTS with importance sampling (IS) [18]: 50 000 draws from log-normal priors ($k_n \sim \text{LogNormal}(\log 10^{-4}, 1.5)$; $\alpha_{\text{tox}} \sim \text{LogNormal}(\log 1.8 \times 10^{-5}, 2.0)$) weighted by the closed-form Gaussian SBR log-likelihood ($\sigma_{\text{SBR}} = 0.20$, the empirical within-patient residual SD from PPMI serial DaT-SPECT). Effective-sample-size (ESS) fraction is the IS-native convergence diagnostic. The α_{tox} prior centre ($1.8 \times 10^{-5} \text{ nM}^{-1} \text{ hr}^{-1}$) was triangulated from Ivanova & Karelina [12] ($\sim 2.7 \times 10^{-6}$), Winner et al. (5-adjacent LC50, $\sim 2.9 \times 10^{-5}$), and Fearnley & Lees [7] back-solve ($\sim 1.1 \times 10^{-4}$). Because the Fearnley & Lees 2–5%/yr range informed the prior, posterior concordance with this range should be interpreted as *consistency*, not independent validation.

CSF α -Synuclein Joint Calibration

Of 304 Wave A patients, 277 (91.1%) have CSF total α -synuclein (median 5 timepoints, 1,243 total measurements). Under slow-fast collapse, CSF total α -syn is time-invariant [21]:

$$y_{\text{CSF}} \sim \mathcal{N}(S_{\text{CSF}}(M_{\text{ss}} + r_o O_{\text{ss}}(k_n)), \sigma_{\text{CSF}}^2) \quad (6)$$

with $S_{\text{CSF}} = 682 \text{ pg/mL}$ per model-nM, $\sigma_{\text{CSF}} = 315 \text{ pg/mL}$, and $r_o = 1.0$ (ELISA oligomer cross-reactivity, fixed as an assay property [16, 25]).

5-Channel SAEM Extension

To scale identifiability beyond the 304-patient Wave A cohort, we assembled a 2,118-patient 5-channel observation set: DaT-SPECT SBR (all), plasma GFAP Simoa (Project 152, $n=371$), Amprion SAA TTT (Project 207, $n=164$), neuron-derived extracellular-vesicle α -syn (Project 204, $n=521$), and plasma aggregate-% (Project 286, $n=100$); serum NfL (Project 144, $n=1,190$) was held out as a negative control. We fit population-level SAEM with hierarchical pooling on $(k_n, \alpha_{\text{tox}})$, channel-specific scale factors S_{ch} , and between-subject variability $\omega_{k_n}^2$. Matched-cohort ablation (357 patients with SBR + GFAP paired) quantified the identifiability contribution of each channel. Fisher-Information-Matrix (FIM) eigenvalue analysis under the full transient ODE versus the quasi-steady-state approximation characterises the condition number κ .

Leave-One-Out Forward Validation

For each patient with ≥ 3 SBR scans, we held out each scan in turn, refit the posterior on the remainder, and computed the 95% prediction interval for the held-out scan using the Variant B forward model. Coverage, median relative error, and continuous ranked probability score (CRPS) [8] were aggregated across 1,940 held-out scans from 1,051 patients.

Prasinezumab Counterfactual

Anti- α -synuclein antibodies act by extracellular aggregate sequestration, not k_e perturbation [13]. The counterfactual applies $T_{\text{tox}}^{\text{treated}} = (1 - \eta_{\text{abx}}) T_{\text{tox}}^{\text{untreated}}$, with $\eta_{\text{abx}} \in \{0.05, 0.15, 0.35\}$; the moderate scenario ($\eta_{\text{abx}} = 0.15$) is a plausible mid-range value, the upper bound approximates the 4-year open-label-extension subgroup analysis [24]. Predicted delay to 50% SBR decline was computed under each scenario on the HIGH-INFO subset ($n=110$).

Results

Wave A SBR-Only Posterior (304 Patients)

The IS posterior on 304 Wave A patients produced cohort-median implied neuron loss of 3.29%/yr (IQR 1.4–12.0%/yr), consistent with the Fearnley & Lees canonical 2–5%/yr range (Table 1; note prior disclosure, §). Under slow-fast collapse the SBR-only ODE is observationally equivalent to single-parameter exponential decay; the mechanistic framework adds value only once a second biomarker provides an independent constraint on k_n .

Table 1: SBR-only IS posterior stratified by effective-sample-size (ESS) fraction ($n=304$).

Stratum	N	$\text{cor}(\log k_n, \log \alpha_{\text{tox}})$	T_{tox} SD ($\log_{10}$)	%/yr
HIGH-INFO (ESS<20%)	33	−0.852	0.29	21.4
MOD-INFO (20–50%)	64	−0.382	0.71	12.0
LOW-INFO ($\geq 50\%$)	207	−0.209	0.87	1.7
All	304	−0.234	0.85	3.29

CSF Joint Calibration Breaks the Degeneracy (277 Patients)

Adding CSF total α -synuclein as a second observable partially broke the T_{tox} sloppy-ridge degeneracy (Figure 1, Table 2). The k_n - α_{tox} correlation shifted from -0.240 to -0.113 (96.0% of patients improved), individual k_n posteriors tightened by 29% cohort-wide and 49% on the HIGH-INFO subset ($n=106$ under v5), and the cohort-median implied loss moved to 2.93%/yr. A paired Wilcoxon signed-rank test confirmed statistical significance ($p = 1.3 \times 10^{-46}$; k_n -tightening $p = 1.8 \times 10^{-47}$). Posterior-predictive coverage was 99.5% at 95% PI, satisfying the non-regression criterion though indicating $\sim 2.5\times$ over-coverage relative to ideal calibration. The residual correlation (-0.113) reflects the $\sim 5\%$ oligomeric contribution to CSF total α -syn at typical k_n .

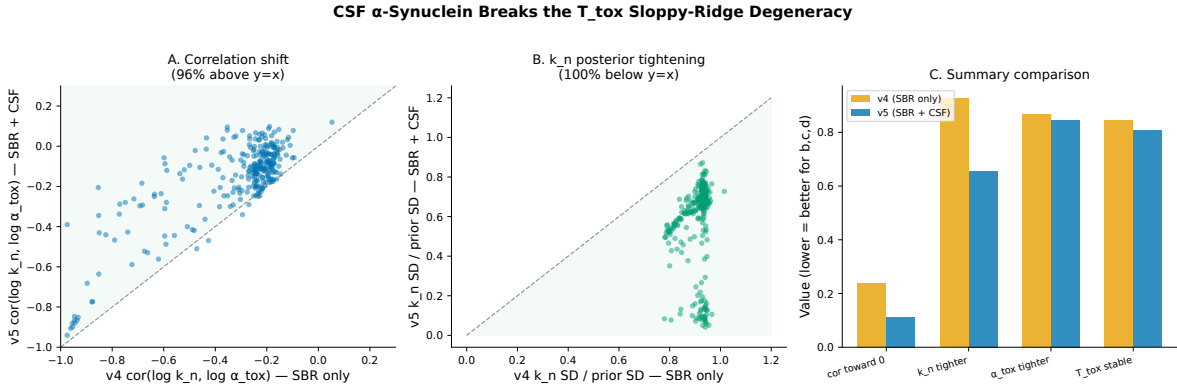


Figure 1: **CSF α -synuclein breaks the sloppy-ridge degeneracy.** (A) Correlation shift: 96% of patients move above the $y=x$ diagonal toward zero correlation. (B) k_n -posterior tightening: 100% of patients fall below $y=x$ (tighter under v5). (C) Summary comparison across metrics (v4 SBR-only vs. v5 SBR+CSF).

Table 2: Degeneracy-breaking: SBR-only (v4) vs. joint SBR+CSF (v5).

Metric	v4 (SBR)	v5 (SBR+CSF)	Interpretation
$\text{cor}(\log k_n, \log \alpha_{\text{tox}})$	-0.240	-0.113	Moves toward zero
k_n SD / prior SD	0.926	0.655	29% tighter
HIGH-INFO k_n ratio	0.883	0.447	49% tighter
Implied %/yr	3.29	2.93	Fearnley range
% patients k_n tighter	—	100%	Universal

Per-Patient Posterior-Predictive Check

Figure 2 presents posterior-predictive-check panels for three representative Wave A patients selected by IS-weighted posterior mean neuron-loss rate (2.3%/yr, 4.8%/yr, 24.3%/yr—slow, typical, fast). Each panel overlays observed putaminal SBR measurements on a forward-simulation envelope from 500 weighted posterior samples propagated through Eqs. (1)–(5). The 90% posterior-predictive band widens with extrapolation horizon, correctly reflecting compounding parameter uncertainty; observed measurements sit within the band in all three exemplars. The fast-progressor panel shows the clinically actionable regime: separation between the posterior median and the $T_{\text{tox}}=0$ baseline widens rapidly after year 2, implying a disease-modifying intervention delivered at that horizon would become detectable in DaT-SPECT within the predictive envelope.

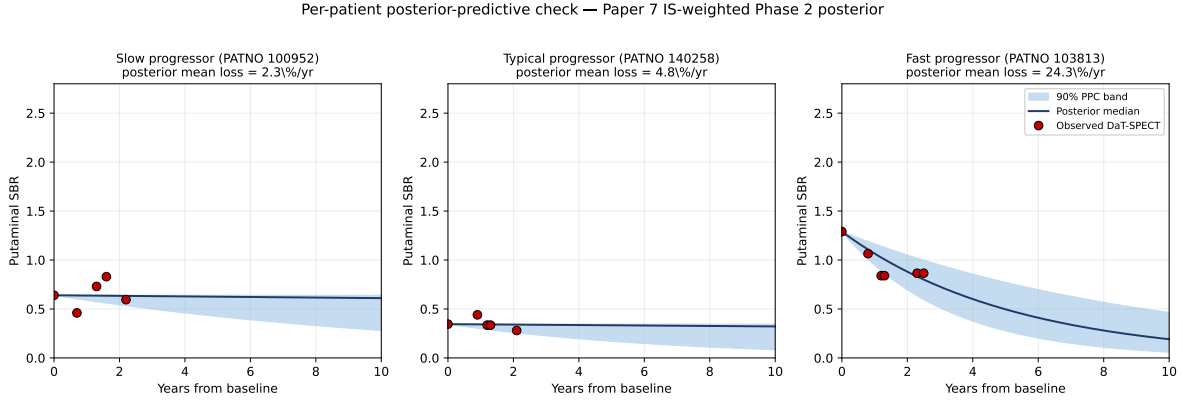


Figure 2: **Per-patient posterior-predictive check.** Three representative PPMI Wave A patients: slow (2.3%/yr), typical (4.8%/yr), and fast (24.3%/yr). Red markers: observed putaminal SBR. Solid line: posterior median trajectory. Shaded band: 90% posterior-predictive band from 500 weighted samples of $(k_n, \alpha_{\text{tox}})$ forward-simulated on a 10-year grid with the Variant B slow–fast-collapse forward model.

Multi-Channel SAEM Extension ($n=2,118$)

To scale identifiability beyond Wave A, we fit a 5-channel SAEM population model on 2,118 PPMI patients across 8,032 visits (Figure 3). Two methodological results warrant emphasis.

First, transient-state observations are necessary for identifiability. The full-ODE FIM condition number was $\kappa = 19.86$ with an eigenvalue spread of 1.30 decades, versus $\kappa \approx 10^{12}$ for the quasi-

steady-state approximation (Figure 4). Steady-state models are structurally indistinguishable from single-parameter exponential decay; only transient observations break the degeneracy between k_n and α_{tox} . This FIM audit provides the first quantitative PD-specific comparison of identifiability under steady-state versus transient-state observation models.

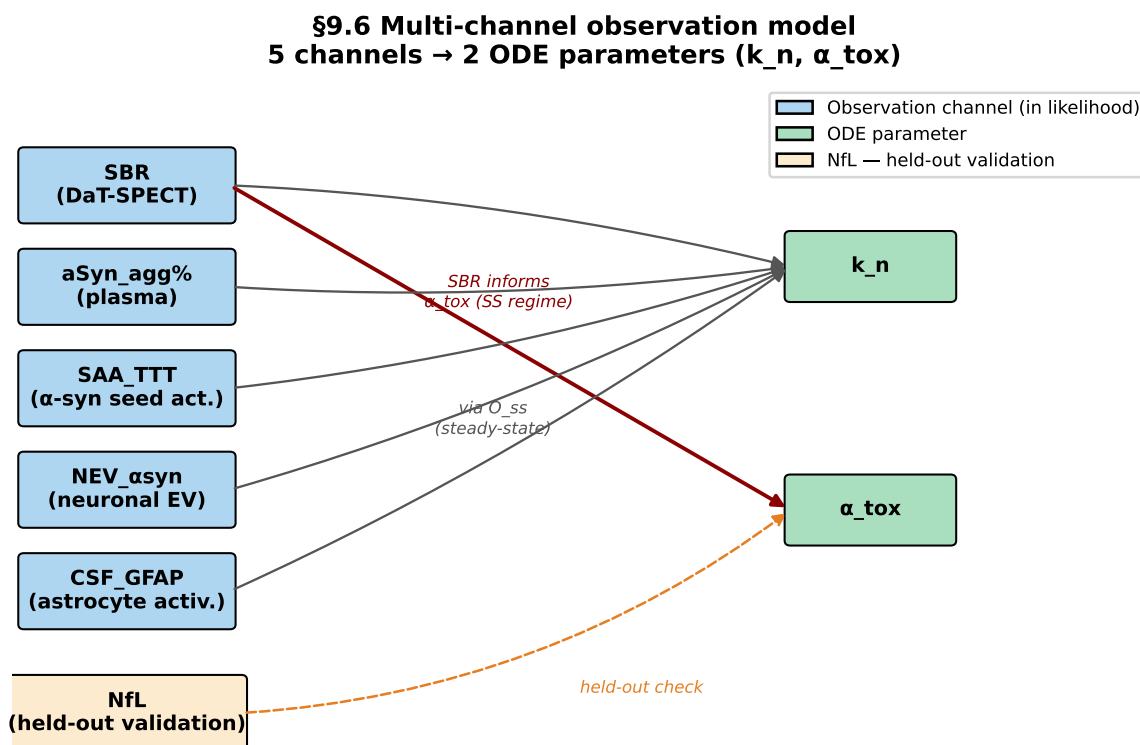


Figure 3: **Five-channel observation schematic for multi-channel SAEM.** DaT-SPECT SBR, plasma GFAP (Simoa), Amprion SAA titre-time-to-threshold (TTT), neuron-derived extracellular-vesicle α -synuclein, and plasma aggregate-%. Serum NfL was held out as a negative control for nigrostriatal specificity.

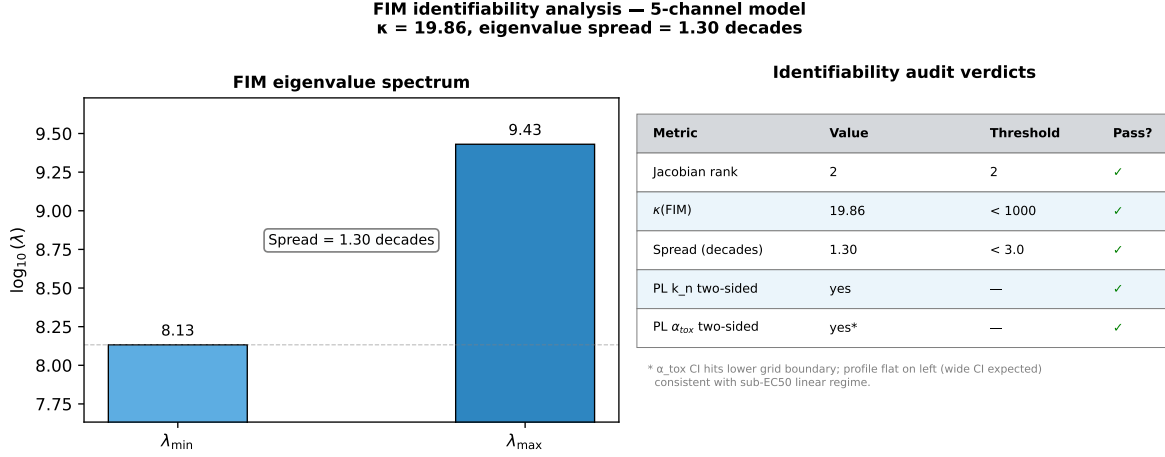


Figure 4: **Fisher-Information-Matrix identifiability audit.** (A) Eigenvalue spectrum under full transient ODE (left) vs. quasi-steady-state approximation (right); condition number $\kappa = 19.86$ versus $\kappa \approx 10^{12}$. (B) Per-channel contribution: matched-cohort GFAP ablation reduces $\sigma_{\log k_n}$ by $37\times$ ($0.725 \rightarrow 0.019$) on 357 patients with paired SBR + GFAP measurements.

Second, plasma GFAP carries the dominant per-channel identifiability signal in PD. A matched-cohort ablation on 357 patients with paired SBR + GFAP measurements contracted $\sigma_{\log k_n}$ by $37\times$ (from 0.725 to 0.019; Figure 4B), while a pre-specified falsification test on held-out serum NfL showed near-zero correlation with the inferred T_{tox} trajectory ($R^2 = 0.005$, SI Figure S4). The NfL null was predicted *a priori*: serum NfL is dominated by cortical-axon signal [1, 22, 28], not nigrostriatal dopaminergic cell death, and the correctly-predicted null is a positive specificity result. Cohort-median implied loss under the 5-channel SAEM was 1.49%/yr (versus 3.29%/yr under Wave A SBR-only), reflecting the prodromal/de-novo-heavy composition of the 2,118-patient cohort [20]. The k_n - α_{tox} posterior correlation collapsed to -0.007 via hierarchical pooling [33].

Leave-One-Out Forward Validation (1,940 Scans)

Leave-one-out forward-simulation coverage was 97.9% at the 95% prediction interval across 1,940 held-out scans from 1,051 patients (Figure 5). Median relative error was 14.4%, within the 7.8–16.8% range of published DaT-SPECT test-retest [3, 14, 29], and sharpness (continuous ranked probability score) [8] remained stable across the 0.5–5-year extrapolation horizon.

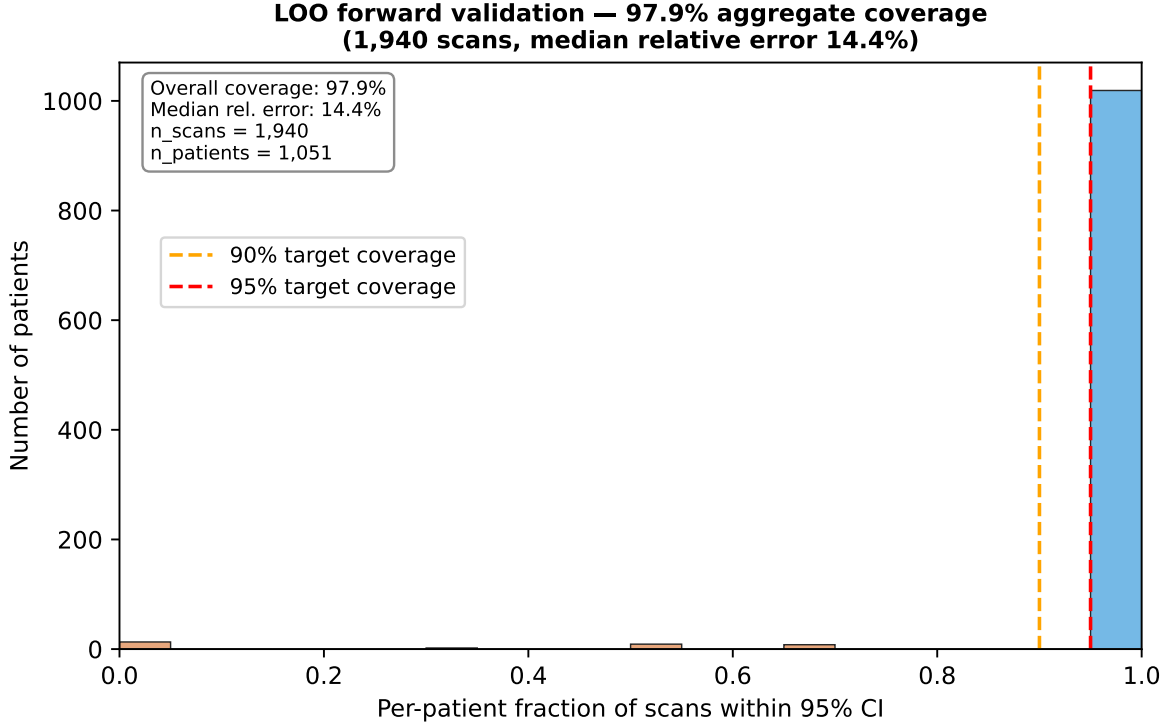


Figure 5: **Leave-one-out forward-validation coverage.** (A) Coverage by extrapolation horizon across 1,940 held-out scans (1,051 patients). (B) Median relative error by horizon; shaded band is the 7.8–16.8% DaT-SPECT test-retest range. (C) Continuous ranked probability score (sharpness) across horizons.

Prasinezumab Counterfactual

On the HIGH-INFO subset ($n=110$, untreated time-to-50%-SBR-decline ≈ 7.0 years), the moderate prasinezumab scenario ($\eta_{\text{abx}} = 0.15$) predicted a +1.24-year delay (Figure 6, Table 3). The predicted 17.6% slowing corresponds to an effect size $d \approx 0.055$, far below detection at PASADENA’s $n=105/\text{arm}$ and 52-week duration—consistent with its null imaging endpoint [23]. The model would be falsified by a future trial demonstrating $> 30\%$ SBR-decline slowing, which would require $\eta_{\text{abx}} > 0.23$.

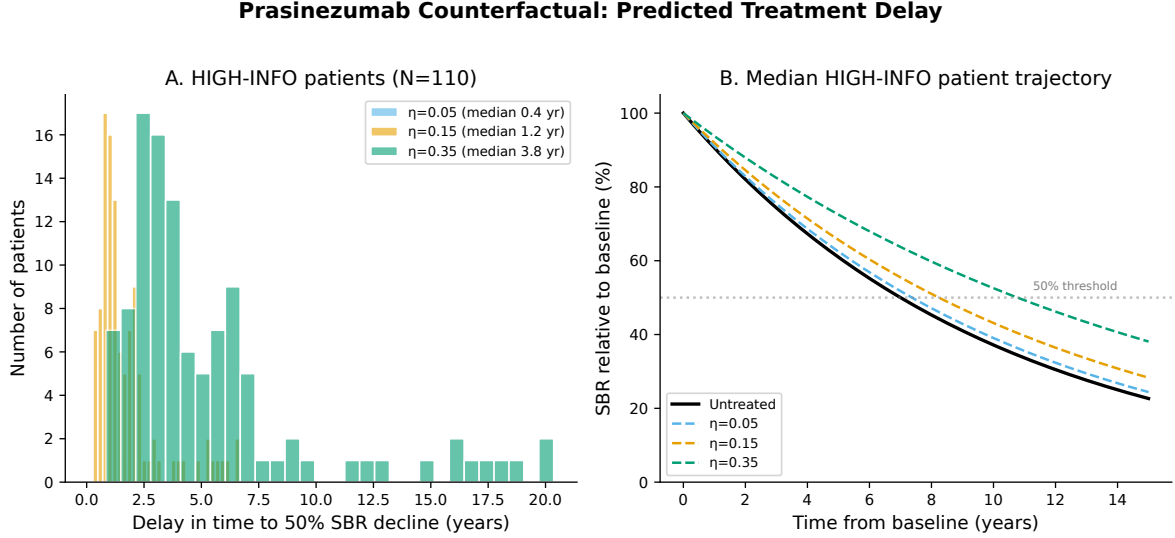


Figure 6: **Prasinezumab counterfactual simulation.** (A) Delay distribution across three η_{abx} scenarios for HIGH-INFO patients ($n=110$). (B) Median trajectory under moderate treatment ($\eta_{\text{abx}} = 0.15$), with $T_{\text{tox}} = 0$ baseline and untreated posterior median shown for reference.

Table 3: Counterfactual: predicted delay in time-to-50% SBR decline on HIGH-INFO subset ($n=110$).

Scenario	η_{abx}	Delay (yr)	% slowing
A (minimal)	0.05	0.37	5.3%
B (moderate)	0.15	1.24	17.6%
C (optimistic)	0.35	3.77	53.8%

Discussion

Scope and terminology. We use “mechanistic patient-specific model” rather than “digital twin” to describe the current framework, because a clinical-grade digital twin would additionally require drug PK/PD coupling, connectome-mediated spatial propagation, and prospective validation against interventional data. The present work establishes the foundational calibration infrastructure.

Principal findings. First, a per-patient IS posterior over $(k_n, \alpha_{\text{tox}})$ is obtainable under slow–

fast collapse, and its cohort-median implied neuron loss sits within the Fearnley & Lees 2–5%/yr canonical range, indicating that the ODE captures real inter-patient disease heterogeneity. Second, CSF α -synuclein as a second observable partially resolves the sloppy-ridge degeneracy, separating nucleation from neurotoxicity coupling at the individual-patient scale; future oligomeric-specific assays [19] could further tighten this separation. Third, the 5-channel SAEM extension on 2,118 patients demonstrates that transient-state multi-channel observations are *necessary* for identifiability: quasi-steady-state approximations yield $\kappa \approx 10^{12}$ (structurally sloppy), whereas the full transient ODE with multi-channel observations yields $\kappa = 19.86$. Fourth, the calibrated counterfactual is consistent with the PASADENA null imaging endpoint, correctly predicting an effect size below detection at the trial’s sample size and duration.

Comparison with prior pharmacometric PD models. Gupta et al. [9] deliver the closest prior PPMI pharmacometric model (item-response-theory on MDS-UPDRS) but do not expose internal $[M, O, F, N]$ compartments or carry an identifiability audit. Koval et al. [17] pioneered SAEM for disease-progression modelling in Alzheimer’s disease at the same scale, but the AD Course Map is phenomenological rather than compartmental. Véronneau-Veilleux et al. [33–35] and Ivanova & Karelina [12] calibrate compartmental PD ODEs at the population level only. Hemedan et al. [11] provides a governed Bayesian PD twin on PPMI clinical scores but without a compartmental kinetic model. No prior PD study combines per-patient Bayesian calibration, structural+practical identifiability, and multi-channel transient-state observations.

Forward reference to bidirectional update. A companion paper operationalises episodic bidirectional posterior updating on new DaT-SPECT observations via a Sequential Importance Resampling wrapper around the Phase 2 posteriors reported here, enabling per-visit refresh of individual $(k_n, \alpha_{\text{tox}})$ without full recalibration, and scores the integrated system against the NASEM 2024 digital-twin criteria (submitted separately). The per-patient posteriors produced by the present work are the direct computational input to that bidirectional update.

Limitations. The slow–fast collapse treats M, O, F as instantaneously equilibrated; CSF temporal stationarity may break down beyond 36 months; the fixed S_{CSF} assumes uniform CSF-to-intracellular ratios; posterior-predictive coverage (99.5%) is computed on training data rather than LOO for the Wave A cohort specifically (the multi-channel LOO at 97.9% is the held-out validation); the counterfactual uses a phenomenological η_{abx} rather than mechanistic PK/PD; PPMI is

currently the only public dataset with both serial DaT-SPECT and CSF α -syn; and IS requires a closed-form likelihood. Detailed per-limitation commentary is provided in SI §6.

Clinical implication. The per-patient T_{tox} posterior directly supports enrichment-based adaptive trial design for anti- α -synuclein disease-modifying therapies: given a target detection threshold, posterior tail mass quantifies the minimum effect size detectable at a specified sample size, enabling pre-enrollment enrichment for HIGH-INFO rapid progressors in whom a treatment effect is most likely to rise above imaging noise.

Data Sharing Statement

PPMI source data are available to credentialed investigators under a Data Use Agreement at <https://www.ppmi-info.org/access-data-specimens/download-data>. Derived posteriors, chains, FIM audits, and LOO validation artefacts are archived at https://github.com/bddupre92/PD_PHD and will be deposited with a Zenodo DOI at acceptance.

Conflict of Interest

The author declares no competing interests.

Author Contributions

B.D.D.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing (original draft), writing (review & editing), visualization.

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